

# -Theory: A Spectral Formalism of Physical Theories Based on the Structure of Operational Distinguishability

## Abstract

We propose a formalism for physical systems based on the structure of operational distinguishability between states. A physical system is represented as a weighted graph of distinguishability, while observables are interpreted as spectral properties of the corresponding operator. We show that quantum mechanics and spacetime geometry can be recovered as limiting cases of this construction. We also discuss a possible correction to interference phenomena related to the informational structure of measurement devices.

## 1 1. Introduction

Modern physics relies on different mathematical frameworks to describe quantum and gravitational phenomena. However, these approaches are based on fundamentally different primitive structures: Hilbert spaces in quantum mechanics and smooth manifolds in general relativity.

In this work, we propose a unified formalism based on the assumption that the fundamental physical structure is not space, fields, or states, but the structure of operational distinguishability between observable states.

## 2 2. Basic Construction

We consider the structure:

$$\chi = (\mathcal{D}, R, W)$$

where:

- $\mathcal{D}$  is the set of operationally distinguishable states,

- $R \subseteq \mathcal{D} \times \mathcal{D}$  is the relation of observed connectivity,
- $W : R \rightarrow \mathbb{R}_{\geq 0}$  assigns weights of distinguishability.

Based on this structure, we define the distinguishability operator:

$$(\hat{\Delta}_\chi f)(i) = \sum_j W_{ij}(f(i) - f(j))$$

### 3 3. Fundamental Principle

Physical observables are interpreted as the spectrum of the distinguishability operator:

$$\text{Physics} \sim \text{Spec}(\hat{\Delta}_\chi)$$

### 4 4. Core Idea

The central idea of -Theory is:

- space emerges from distinguishability structure;
- dynamics arises from the redistribution of weights  $W$ ;
- quantum effects correspond to spectral interference modes;
- measurement affects the structure of distinguishability, not only the system state.

### 5 5. Relation to Quantum Mechanics

In the continuum limit, weights are interpreted as local contributions to the action:

$$W_{ij} \rightarrow \mathcal{L}(x, \dot{x})\Delta t$$

leading to the standard action:

$$S = \int \mathcal{L}(x, \dot{x})dt$$

and the quantum amplitude:

$$A = \sum_{\text{paths}} e^{iS/\hbar}$$

Thus, quantum mechanics arises as a limiting spectral regime of the -structure.

## 6 6. Geometric Limit

The distance between states is defined as:

$$D(i, j) = \min_{\gamma} \sum_{(k, l) \in \gamma} W_{kl}$$

In the continuum limit, the operator  $\hat{\Delta}_{\chi}$  converges to the Laplace–Beltrami operator on a curved manifold:

$$\hat{\Delta}_{\chi} \rightarrow \square_g$$

This links distinguishability structure with spacetime geometry.

## 7 7. Main Theorem

**Theorem (-representability of physical theories).**

Any physical theory  $T$  admitting an operational description in terms of observables can be represented as a spectral structure of the distinguishability operator:

$$T \equiv [(\mathcal{D}, R, W), \text{Spec}(\hat{\Delta}_{\chi})]$$

### 7.1 Proof idea

1. A physical theory defines a state space  $\mathcal{S}$ .
2. Define observational equivalence  $x \sim y$ .
3. Construct the quotient space  $\mathcal{D} = \mathcal{S} / \sim$ .
4. Dynamics induces  $R$  and weights  $W$ .
5. The operator  $\hat{\Delta}_{\chi}$  is defined on this structure.
6. Its spectrum determines observable quantities.

## 8 8. Testable Consequence

-Theory allows for a possible dependence of interference effects on the measurement device structure:

$$\Delta\phi = \Delta\phi_{QM} + \epsilon \cdot \Omega_{\text{det}}$$

where  $\Omega_{\text{det}}$  characterizes the informational complexity of the detector. This implies:

$$\frac{\partial I}{\partial \Omega_{\text{det}}} \neq 0$$

## 9 9. Discussion

The proposed formalism does not introduce new fundamental entities but reinterprets existing physical theories as spectral manifestations of distinguishability structure.

In this sense, quantum mechanics and spacetime geometry appear as different limiting regimes of a single mathematical construction.

## 10 10. Conclusion

-Theory provides a unified spectral formalism in which physical theories can be reconstructed from the structure of operational distinguishability. The framework allows for a testable deviation from standard quantum mechanics related to the structure of measurement.

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